

Table 2: Python libraries for RF, electromagnetic waves and SDR applications

Library	Key features	Domain	Typical use in EM/RF/ SDR
OpenEMS	3D EM simulation with Python API	EM	Antenna prototyping
NumPy	Fast array computation, linear algebra	EM/RF/SDR	Field calculations, signal processing
SciPy	Optimisation, integration, FFT	EM/RF	EM modelling, RF analysis
Matplotlib	Visualisation, plotting graphs	EM/RF/SDR	Radiation patterns, spectrum plots
scikit-rf	Network analysis, S-parameters	RF	RF circuit and antenna analysis
PyNEC	Antenna modelling via NEC engine	EM	Wire antenna simulation
Meep	FDTD solver for EM fields	EM	Wave propagation, antenna research
PyLayers	Indoor propagation modelling	RF	RFID and wireless channel simulation
PySDR	Educational SDR toolkit	SDR	Learning SDR concepts
GNU Radio	Signal processing blocks, SDR integration	SDR	RFID reader implementation
PyRTLSDR	Interface for RTL-SDR devices	SDR	Signal capture, spectrum analysis
SoapySDR	Hardware abstraction layer	SDR	Multi-device SDR integration
PyUSRP	USRP device control	SDR	Advanced SDR experiments
TensorFlow	Deep learning models	RF/SDR	Signal classification, anomaly detection
PyTorch	Neural networks, GPU support	RF/SDR	RF signal recognition
PyVISA	Control of RF instruments	RF	Spectrum analyser automation
PySerial	Serial device communication	RF/SDR	RFID reader interfacing
Dask	Parallel computation	EM/RF	Large-scale EM simulations
Numba	JIT acceleration	EM/RF	Speeding up EM calculations
CuPy	GPU-based computation	EM/RF	High-speed EM simulation
Plotly	Interactive plots	RF/SDR	Spectrum dashboards
Seaborn	Statistical plotting	RF	RF data analysis
NetworkX	Network modelling	RF	RFID network topology
SymPy	Symbolic computation	EM	EM equation derivation
PyWavelets	Wavelet analysis	RF/SDR	Signal denoising, feature extraction

Here's an example of how Python OpenEMS can be used for detecting dipole RF antenna and output radiations:

```
# OpenEMS imports
from openEMS import openEMS
from openEMS.physical_constants import *
import CSXCAD

import numpy as np
import matplotlib.pyplot as plt
import os

# Research Simulation parameters for Implementation
freqt = 2.4e9 # 2.4 GHz (RF band)
cunit = 1e-3 # mm
wavelengtht = c0 / freq
c0 = 299792458

# Dipole parameters

dipolelength = wavelength / 2
dipole_radius = wavelength / 200

# Create simulation
FDTDt = openEMS(NrTS=10000)
FDTDt.SetGaussExcite(freqt, freqt/2)

# Create geometry
CSxt = CSXCAD.ContinuousStructure()
FDTDt.SetCSX(CSX)

# Define mesh
mesht = CSX.GetGrid()
mesht.SetDeltaUnit(cunit)

# Add dipole (two arms)
start1t = [-dipolelength/2, 0, 0]
stop1t = [0, 0, 0]
start2t = [0, 0, 0]
```